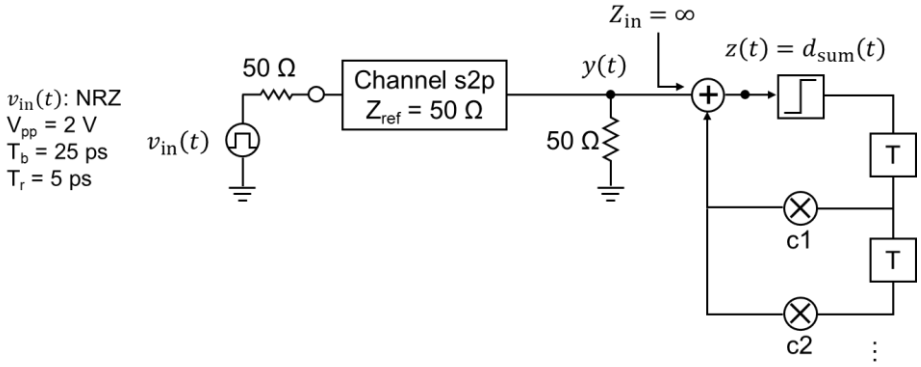
Homework: DFE simulation and design

In the previous homework, we have designed CTLE and FFE for a 50cm strip line channel. In this homework, we are going to design a DFE for the same channel (so please use the same s2p). Same parameters as before (40 Gbps NRZ, risetime 5 ps, impedance matching, and that the transfer function from $V_{\text{in}}(f)$ to $y(f)$ is $H = S_{21}^{\text{Ch}}/2$).



You have to determine the parameters for the DFE yourself. In particular,

1. The DFE tap coefficients. You can use the method discussed in the class note.
2. The threshold voltage Vt. You can manually tune it to find the best value.
3. The sampling point (sample_pt). You can manually tune it to find the best value.

For *each* of the following cases, show the post-DFE eye diagram, and a 1-ns snapshot of the post-DFE waveform versus the post-channel waveform.

1. One DFE tap.
2. Two DFE taps.
3. Three DFE taps.

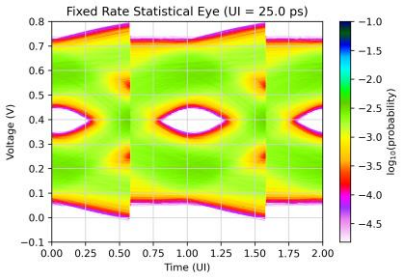
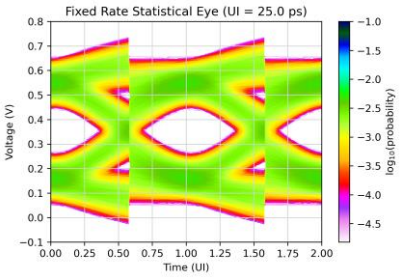
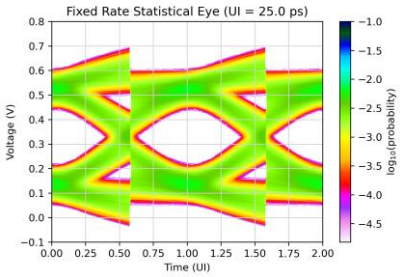
Comment on the following questions:

1. How to determine a good Vt automatically (without tuning)?
2. How to determine a good sampling point automatically (without tuning)?
3. Based on your snapshots of the waveform, does your DFE always make correct decisions? If not, point out an example where wrong decision was made.

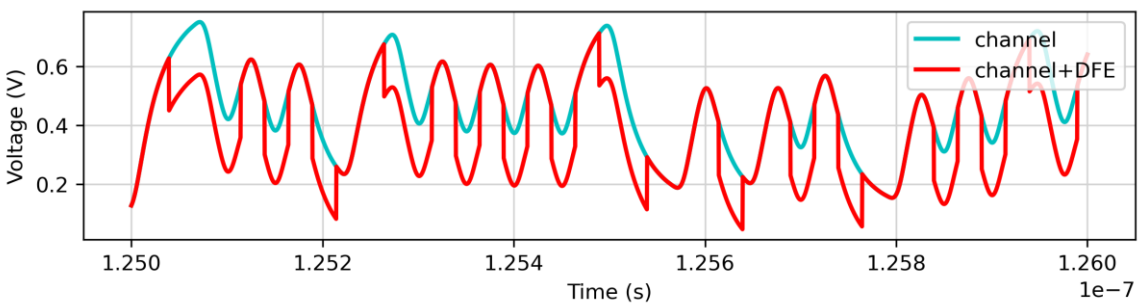
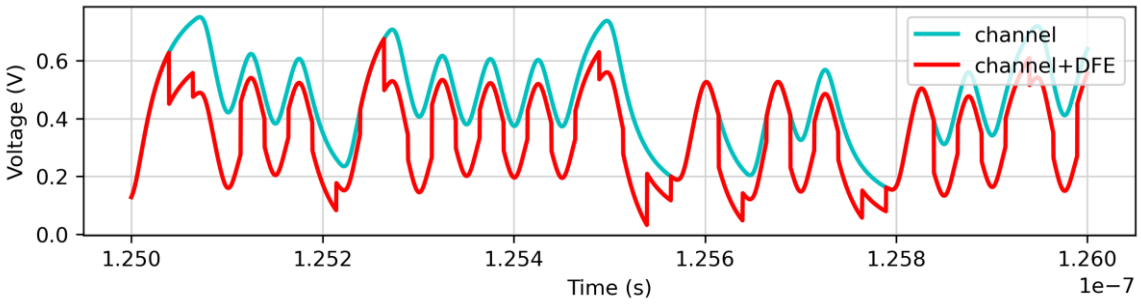
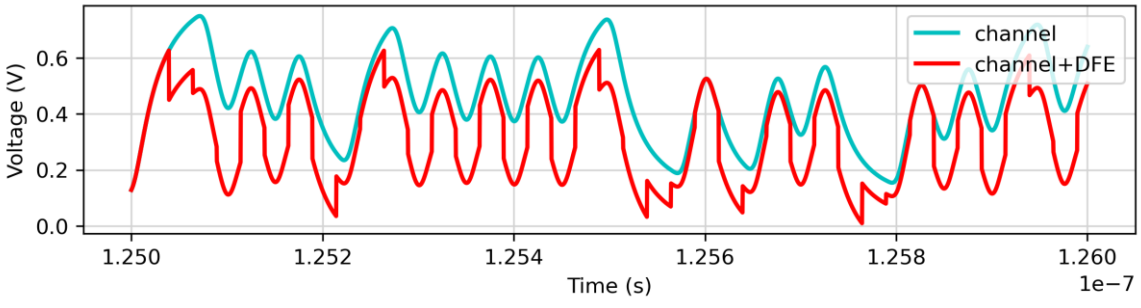
Optional problem: Implement your DFE in ADS ChannelSim to test its performance. See appendix.

Note: The DFE simulator code of this homework is *real* simple. The main problem that you need to think is how to choose the Vt and sampling point automatically.

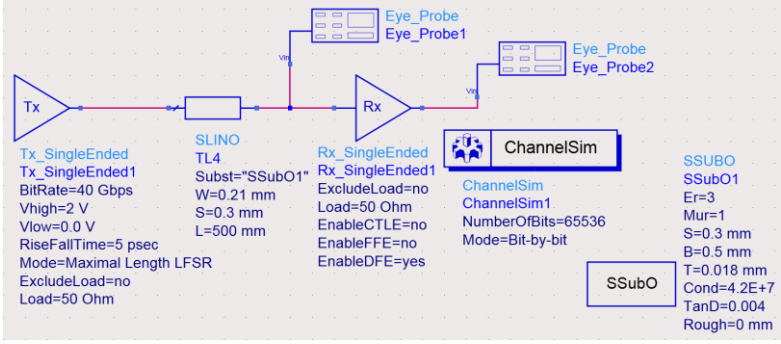
Appendix 1: Example answer

One tap	Two taps	Three taps
		

Waveforms

One tap	
Two taps	
Three taps	

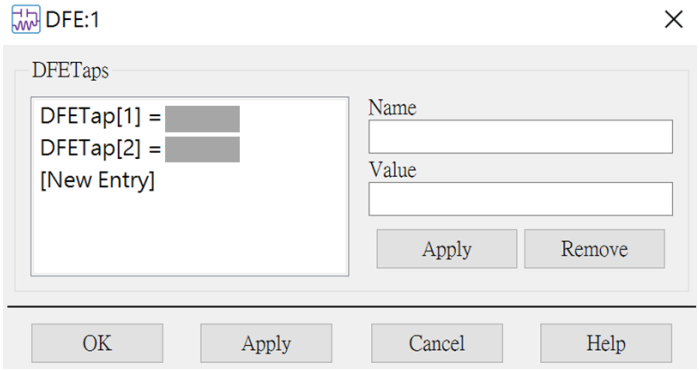
Appendix 2: ADS setting



Double click the Rx to set the DFE. Set Slicer output to 1/0.



Click “Edit” to set the DFE tap coefficients.



ADS’s post-DFE eye and post-DFE vs post-channel waveforms

